View-Semantic Transformer With Enhancing Diversity for Sparse-View SAR Target Recognition

Zhunga Liu, Feiyan Wu, Zaidao Wen, *Member, IEEE*, and Zuowei Zhang

***Abstract*— With the rapid development of supervised learning-based synthetic aperture radar (SAR) target recognition technology, it is easy to find that the recognition performance is proportional to the number of training samples. However, the biased data distribution and under-representation of the model caused by incomplete data within categories exacerbate the challenge of SAR interpretation. In this article, we propose a new view-semantic transformer network (VSTNet) that generates synthesized samples to complete the statistical distribution of training data and improve the discriminative representation of the model. First, SAR images from different views are encoded into a disentangled latent space, which allows us to synthesize data with more diverse views by manipulating view-semantic features. Second, the synthesized data as a complement effectively expands the training set and alleviates the overfitting problem of limited data in sparse views. Third, the proposed method unifies SAR image synthesis and SAR target recognition into an end-to-end framework to boost their performance against each other. Experiments conducted on moving and stationary target acquisition and recognition (MSTAR) data demonstrate the robustness and effectiveness of the proposed method.**

***Index Terms*— Incomplete data, sparse views, synthetic aperture radar (SAR) target recognition, view-semantic transformer.**

I. Introduction

SYNTHETIC aperture radar (SAR) imaging technology is essential in civil and military applications. The maturity of this technology, accompanied by the large amounts of high-resolution SAR image data, has resulted in significant progress in the field of SAR target recognition [1], [2]. However, due to the characteristics of SAR imaging, the SAR data will show specific differences when the SAR sensors are planned at different views. In addition, the collected SAR data exhibits insufficient intraclass diversity (incomplete data) due to the sparse views of the SAR sensors, as is shown in Fig. 1, which poses a further challenge to SAR interpretation or automatic target recognition [3], [4]. Therefore, many efforts have been made in recent years to make a breakthrough in SAR target recognition with incomplete data.

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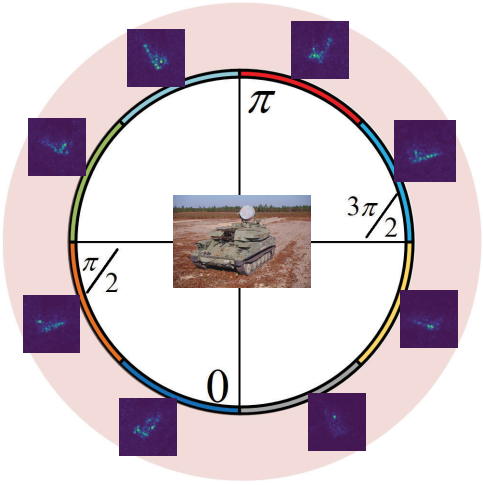


Fig. 1. SAR sensors obtain high-resolution images of objects from different views.

Data synthesis is a straightforward method to mitigate the sensibility to distribution shift caused by incomplete data when generating complementary training data [5]. There are three kinds of related research on SAR data generation. The traditional SAR data augmentation operation is to build a model of the electromagnetic characteristics of the target, which can ensure the reliability of the generated data by parameterizing the target from the perspective of physical modeling. Still, the disadvantage is that it needs a lot of computing resources [6], [7], [8]. With the growth of computing resources in recent years, there has been a great success in generating high-quality SAR data [9], [10], [11], [12]. Most directly refer to the network structure that performs well in optical image synthesis, so there is little relevance for SAR data [13]. Data augmentation focuses on generating data to improve overfitting in classification by applying label-preserving linear transformations to the original data [14]. However, the biased data distribution is the essence of the overfitting problem. These data synthesis methods do not improve the data distribution determined by higher-level features. The main reason is that there is no analysis and mining of the characteristics of the training data, which has the disadvantage of being blind. In addition, the data generation and the recognition model in the above method are two independent subnetworks that fail to achieve end-to-end optimization in an adjustable feedback manner. The generative model may have an ill-posed solution due to the limitation of incomplete data, and the recognition model trained by synthetic samples must be biased.

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Transfer learning is another practical approach that transfers valuable knowledge from the source domain to the target domain. The common wisdom is that other domain data shared with the SAR domain is transferred to SAR domain-related tasks to help improve model effectiveness [15], [16], [17], [18], [19]. It eliminates the network's dependence on good generalization performance on a large amount of training data. Specifically, these methods used the parameters learned from others to initialize the network of target tasks, which achieved the transfer of generic knowledge to an SAR target recognition task. Among them, the reason for supporting knowledge transfer is that the features extracted by the deep network pay more attention to the high-level semantic features, which ignore the differences between data in different domains. It is worth noting that these methods help solve SAR interpretation problems with incomplete data. However, regarding the scattering characteristics of SAR data, the data used for training the pretrained network is less correlated with them. Therefore, the reason target domain tasks benefit from transfer learning is still ambiguous and needs to be explored.

Multiview learning is specially invented to improve the under-representation of incomplete data by exploiting the consensus and complementary information between different views of the target image. First, a multiview deep-learning framework was proposed to fuse features from multiple views to enable accurate recognition [20]. By contrast, richer feature information obtained from different views improves recognition performance. Many methods based on the traditional sparse representation classification (SRC) are present to explore correlations among multiview SAR images [21]. In addition, long short-term memory (LSTM) based on the self-attention mechanism is widely used in multiview SAR target recognition [22]. In conclusion, multiview strategies generally outperform single-view approaches. However, existing multiview approaches still fail when SAR sensors are planned for sparse-view scenarios. Analytically, collecting complete SAR data is indispensable to obtaining a multiview model with excellent performance. Thus, there is a need to design an SAR target recognition algorithm to supplement existing ones and eliminate the view limitation.

In this article, we propose a view-semantic transformer network (VSTNet) for sparse-view SAR target recognition. Specifically, based on the linear separation assumption of the deep feature space [23], we design a specialized disentanglement module to capture meaningful transformations, including view-semantic factors and others, as much as possible. Furthermore, with the view propagation ability of the semantic augmentation module, a smoother view-semantic representation can be obtained in the feature space. Based on the above-hidden space sampling, the biased distribution decision boundary caused by incomplete data has been alleviated. In this way, a recognition model trained through discriminative representation learning with inputs sampled from the augmented feature space can generalize to a wide range of practical target recognition scenarios.

We summarize our contributions as follows.

1) We propose a new SAR target recognition method termed VSTNet to improve the performance of target recognition. The method draws on the advantages of deep generative models to reconstruct the underlying true distribution of sparse-view training samples, that is, approximately complete data. At the same time, the recognition model is trained by sampling from the calibrated complete distribution. The two modules are nested within each other to facilitate end-to-end optimization. Therefore, the proposed VSTNet overcomes the over-fitting caused by sparse-view data and significantly improves the performance of target recognition.
2) We propose a semantic transformer network that models view-semantic variation factors and then view semantic propagation can be achieved by traversing the variation direction. By doing so, the biased data distribution due to sparse views is compensated, that is, the diversity is enhanced. The proposed network is associated with the recent development of semantic image editing and draws on its advantages in data generation. The key is to search for a direction adversarially in which the view-related feature changes smoothly. By traversing this direction, we will achieve "view propagation" that preserves its identity while obtaining the approximately complete target images.
3) We propose a joint representation learning strategy that utilizes the cooperation and interaction between semantic data augmentation and discriminative feature learning to promote each other. On the one hand, we design a view semantic augmentation module to generate augmented data so that inputs can be sampled from the compensated distribution to improve target recognition performance. On the other hand, directly connecting the discriminant representation to the augmented feature space uses the recognition task to guide the latter to generate augmented features that are beneficial to a reasonable decision boundary.

The rest of this article is organized as follows. Section II reviews some related work associated with various recent development in exploiting the semantic direction of latent space, few-shot learning in recognition, and multitask learning. In Section III, the proposed VSTNet is derived in detail. The extensive experiments will be carried out in Section IV to demonstrate the effectiveness and superiority of the proposed method for SAR target recognition. Section V concludes this article and shows some possible future directions.

II. Related Work

In this section, we introduce the literature from three perspectives as the proposed method for SAR target recognition algorithms, involving semantic image editing in latent space, few-shot learning, and multitask learning. In particular, we highlight the fundamental differences between our method and the other closely related works.

A. SAR Data Synthetics With Generative Models

Generative models aim to estimate the probability distribution of underlying data so that we can sample from a low-dimensional space to generate reliable augmented data.

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Existing data-generation methods can be divided into variational autoencoders (VAEs) [24] and generative adversarial networks (GANs) [25], [26]. VAEs and variants primarily comprise an encoder network and a decoder network; the former maps data points into a latent space, while the latter reconstructs the original input data. GANs and variants implicitly model the underlying distribution of the data with game theory, which is characterized by adversarial learning between the generator and the discriminator. Despite these successes, model performance relies on large amounts of training data to generate new composite data. Therefore, researchers are increasingly exploring solutions to practical problems: how to effectively expand training data to alleviate the impact caused by insufficient and incomplete data.

The latent space for generative models often contains many vectors with semantics that correspond to some semantic transformation in the image space [27], [28]. Furthermore, we find that the control of semantic vectors in latent feature space does change the properties of image space; this property satisfies our requirement of extending incomplete data. The methods for obtaining interpretable semantic directions can be summarized in two ways: applying explicit manual annotations in a supervised manner [29] or solving the optimization issues related to the latent space unsupervisedly [30], [31]. Our proposed method aims to find a view semantic direction in an adversarial manner in which the view-related feature varies smoothly. The difference between our method and others is that we do not only pursue high-resolution generated images, but also pay more attention to alleviating distribution shifts in the augmented feature space and finally obtaining a reasonable decision boundary.

B. Few-Shot Learning

Regarding the perspective of addressing the few-shot learning problem, existing algorithms can generally be divided into three categories. The first category of methods aims to increase the discriminability of the feature representations, such as deep metric learning that produce discriminative features with large interclass and small intraclass distances in the embedding space [32], [33]. The second common category of algorithms seeks to explore the general feature representation of the object such that it can be fit quickly with a small number of samples. With the rapid development of meta-learning theory, the dilemma of few-shot target recognition is greatly alleviated [34], [35]. The last category of methods produces more samples with data augmentation and integrates strong invariance into the learning of robust classifiers [36], [37], [38]. In this article, we are committed to exploring the latent representation of deep generative models to solve the lack of interclass diversity caused by samples with sparse views. In the meantime, connect it with downstream target recognition tasks to realize end-to-end optimization. In addition, joint learning targets the generation of examples that are useful for helping reach sharper classification boundaries.

C. Multitask Learning

Multitask learning (MTL) aims to leverage cooperative or auxiliary relationships between multiple related tasks so that each can benefit from the other [39]. The learning paradigm efficiently uses data and improves network learning efficiency, which surprises some data-starved applications and shows good performance [40]. The critical point in MTL is to correctly balance the relationship between multiple tasks, which is also the future research direction proposed in the conclusion of [40]. In particular, the improper optimization method used on multitasks leads to poorer performance than single-task models. In this case, the key to multitask learning is to confirm what and how information is shared between multitasks. The existing main multitask learning optimization methods include loss weighting [41] and multiobjective optimization [42]. The former is the typical approach to balancing the importance of individual loss functions among tasks. The latter further considers the conflict loss function in multitask optimization. However, the emphasis of the above multitask optimization method is not consistent with that in this article.

Inspired by the curriculum learning [43], the learning process is simulated when humans attempt to learn concepts by gradually increasing the complexity of tasks. Specifically, we propose a target recognition method based on multitask learning. The method enlarges the diversity of the target aspect angles with the help of a generative model, which generates more new samples and improves the performance of target recognition with target samples at limited aspect angles. As we know, the convergence rate of the generative model and the discriminant model is not consistent due to the difference in their theoretical characteristics. On the other hand, the performance of target recognition depends on the augmented feature space learned by the generative model. Therefore, the optimization of multitask learning in this article is closer to the theory of bilevel optimization and task-feedback learning. In this way, the insufficient sample size in discriminant tasks can be alleviated through the solution space of the generative model. Inspired by the task-feedback learning [44], [45], we proposed a joint learning strategy where the generative model and discriminant tasks can cooperate with each other.

III. View-Semantic Transformer Network

In this section, we first introduce the framework of the proposed VSTNet shown in Fig. 2, which simultaneously performs SAR image synthesis and discriminative representation learning for SAR target recognition. Then we go over the method's structure in detail. Finally, we compare our model with several related methods.

A. Overview

We propose a VSTNet that unifies SAR image synthesis and SAR target recognition into an end-to-end learning framework. The structure of the proposed model consists of an encoder E, a generator G, a discriminator D, and a classifier C. In particular, we first feed two SAR images with the same category but different views into encoder E to extract view-related features. The discriminator D is trained to constrain the latent space to close the prior distribution. We use a uniform distribution as the prior for the latent space, and the discriminator constrains the output of the encoder to fill the latent space.

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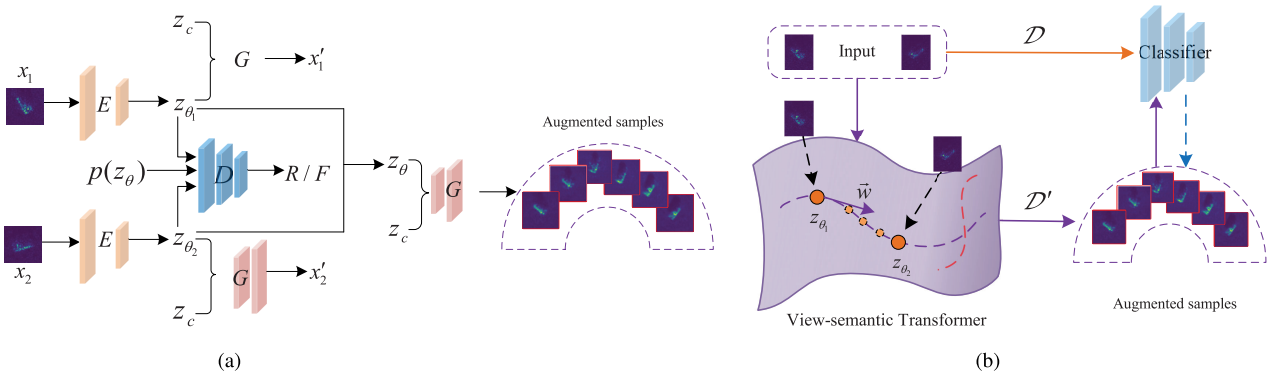


Fig. 2. Overview framework of VSTNet. (a) SAR image synthesis based on disentanglement. (b) SAR target recognition.

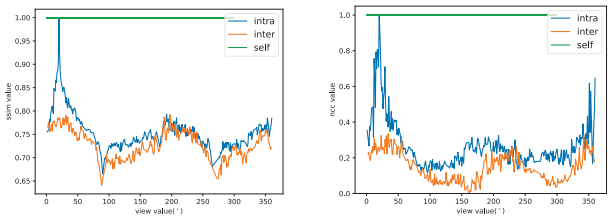


Fig. 3. Correlation curve of target images from different views. Blue line: a ZSU234 SAR image at 20° with the same class from 0° to 360°. Red line: a ZSU234 SAR image at 20° with a 2S-category from 0° to 360°. Green line: the self-correlation of a ZSU234 SAR image at 20°. Correlation measured by (a) structure similarity index measure (ssim) and (b) normalization cross correlation (ncc).

This behavior smooths the distribution of the feature space and avoids the disadvantage of directly using discrete views as supervision information in sparse views. Then, generator G uses the encoded view features and class attributes as inputs to reconstruct the original input. Because of the characteristics of the smooth latent space, view-semantic transformation can be performed to obtain augmented samples complementary to the training data. Finally, a robust recognition model C can be obtained by sampling from the original training data and using augmented samples as input.

B. SAR Image Synthesis and Disentanglement Representation

1) Disentanglement of the View Factor: One of the unique characteristics of SAR imaging is that SAR images of different targets with similar views have a higher correlation. In Fig. 3, we plot the correlation curve between a ZSU234 SAR image at 20° with the same category and a different category from 0° to 360°. The correlation curve shows that the characteristics of an SAR target remain stable when the view varies a small range (≤5°). This presents significant challenges to SAR target recognition with sparse views. We propose a view factor disentanglement-based SAR image synthesis method to address this challenge. A complete multiview SAR dataset can be generated and then applied to train recognition algorithms. Synthetic data can help overcome the challenge of limited data availability and improve the generalizability of the algorithms, allowing them to perform well on previously unseen real-world images. Most importantly, the synthetic data can effectively complement the training data and improve the performance of recognition algorithms.

Disentanglement is a technique used in machine learning to decompose the input images into independent attribute factors. These factors represent specific interpretable concepts, such as the view or semantic information of the image in the feature space. Disentanglement aims to separate these factors so that the more interesting ones can be analyzed or modified independently without affecting each other. In sparse views, disentanglement can separate the view-related attribute from the mixed semantic information of the image. The means of separation allow them to focus on the most relevant information and provide potential benefits for data augmentation. We propose using an encoder–decoder structure with an adversarial learning strategy to achieve disentanglement in SAR images. The proposed method allows us to learn a disentangled representation of the input images that preserves the view-semantic factors and then use this representation to generate synthetic images that preserve the view-semantic factors of the original images.

According to Fig. 2, the encoder E maps two SAR images x_1, x_2 with the same category and different views into the latent space to extract view-related feature $z_{\theta_1}, z_{\theta_2}$. The generator G takes the encoded view-related features and category labels z_c as input, to reconstruct original inputs x'_1, x'_2. Simply, the process can be expressed as $x_1 \to [z_{\theta_1}, z_c] \to x'_1$ and $x_2 \to [z_{\theta_2}, z_c] \to x'_2$. The derivation expression is as follows:

$$z_{\theta_1} = E(x_1), \quad z_{\theta_2} = E(x_2) \tag{1}$$

$$x'_1 = G\big([z_{\theta_1}, z_c]\big), \quad x'_2 = G\big([z_{\theta_2}, z_c]\big). \tag{2}$$

To address the issue of nonsmooth low-dimensional feature space resulting from using sparse view labels as supervised information to estimate view-semantic features, the proposed method uses a discriminator D to constrain the distribution of view features z_{θ_i} (mapped by the view encoder E) to be close to the prior distribution. The uniform distribution as a prior helps to constrain the encoded view factor evenly fill the latent feature space. For the incomplete training data $\mathcal{D}$, we use $p_{\mathcal{D}}(x)$ to denote the original data distribution. The aim of adversarial training is to enforce that the output distribution $q(z_{\theta_i}|x)$ of the encoder E is closed to the prior distribution $p(z_\theta)$. Assuming $p(z_\theta)$ is an uniform distribution, and $z_\theta \sim p(z_\theta)$ denotes sampling from the prior distribution.

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The encoder E and discriminator D are trained with the following objective function:

$$\mathcal{L}_d = \max_D \mathbb{E}_{z_\theta \sim p(z_\theta)} \left[\log D(z_\theta)\right] \tag{3}$$

$$z_{\theta_i} = E(x_i) \tag{4}$$

$$\mathcal{L}_e = \min_E \mathbb{E}_{z_{\theta_i} \sim q\left(z_{\theta_i} \middle| x_i\right)} \left[\log\left(1 - D\left(z_{\theta_i}\right)\right)\right]. \tag{5}$$

For the two disentangled factors, z_c preserves the high-level category-specific information of the target image, while z_{θ_i} represents the view-related features. We concatenate z_c and z_{θ_i} to synthesize target images through the generator G so that the learned representation can explicitly separate the independent factors from the extracted mixed features. Consequently, we formulate the following objective function:

$$\mathcal{L}_r = \mathbb{E}_{x_i, y \sim p_{\mathcal{D}}(x)} \left[\left\| x_i - G\left(\left[z_{\theta_i}, z_c\right]\right)\right\|_2\right] \tag{6}$$

where $\|\cdot\|$ encourages a low L_2 error between the reconstructed target image and the input.

2) View-Semantic Transformer: Deep convolutional neural networks convert the manifolds of high-dimensional images into a linear subspace. This assumption led us in several meaningful directions, corresponding to semantic transformations in the input space. This article uses a linear subspace to achieve data augmentation for sparse views. The aim is to improve the semantic feature smoothness in the deep feature space by performing transformations along meaningful directions and traversing semantic manifolds. Therefore, we conduct data augmentation using the VSTNet to navigate the linear subspace, as is shown in Fig. 2(b). In this way, we traverse the smooth direction of view-related features that can change the "view attribute" without affecting the "category attribute." In other words, we can alter the view semantic by linearly moving the latent code z_{θ_i} along the smooth direction $\vec{w}$ of view features, where α determines the scale, with large α values corresponding to a greater degree of transformation. The generator G takes the "new" view-related features z_θ and category labels z_c as input, to generate augmented data x_{aug}. We formulate the following objective function:

$$\vec{w} = z_{\theta_2} - z_{\theta_1} \tag{7}$$

$$z_\theta = \alpha \cdot \vec{w} + z_{\theta_1}, \quad \alpha \in [0, 1] \tag{8}$$

$$G([z_\theta, z_c]) = x_{\text{aug}}. \tag{9}$$

Since training data with sparse views makes the model vulnerable to the model collapse problem, we avoid it by adding a regularization loss to constrain the generator to generate dissimilar images during training. Referring to existing research, the mode-seeking regularization proposed in Mao et al. [46] is adopted in our method. The diversity loss L_{div} is as follows:

$$\mathcal{L}_{\text{div}} = \frac{\left\|z_\theta - z_{\theta_i}\right\|_1}{\left\|x_{\text{aug}} - x_i'\right\|_1 + \varepsilon}. \tag{10}$$

C. SAR Target Recognition

To deal with the under-representation caused by the sparse views in the training data, we propose to take advantage of generative models to synthesize samples $\mathcal{D}'$ as a supplement to the skewed data distribution. The acquisition of complete training data, which can be described as $D \cup D'$, is a good preparation for training a robust classifier. We calculate the difference between the true class probability y and the estimated output probability $p(y|x)$ or $p(y|x_{\text{aug}})$ stochastically for sample x as well as the augmented sample x_{aug} using the cross-entropy regularization loss L_c and L_c^{aug} as

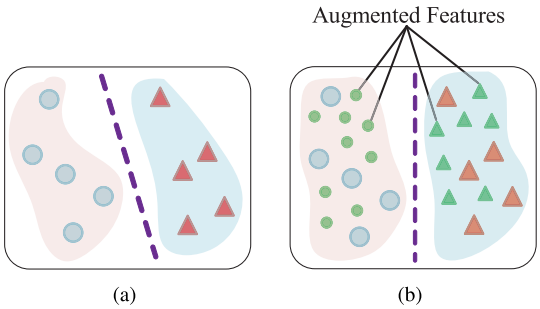


Fig. 4. Visualization of decision boundaries. Visualization result of (a) general representation learning under sparse views data and (b) semantic augmentation and joint representation learning combined under sparse views data.

$$\mathcal{L}_c = \mathbb{E}_{x, y \sim P_{\mathcal{D}}(x)} \left[-y \log P(y \mid x)\right] \tag{11}$$

$$\mathcal{L}_c^{\text{aug}} = \mathbb{E}_{x_{\text{aug}}, y \sim P'_{\mathcal{D}}(x_{\text{aug}})} \left[-y \log P\left(y \mid x_{\text{aug}}\right)\right]. \tag{12}$$

For more stable learning of the classifier, the feature matching loss [47] L_{f_i} and $L_{f_{\text{aug}}}$ is used to limit the output of the intermediate layer

$$\mathcal{L}_{f_i} = \mathbb{E}_{x, y \sim P_{\mathcal{D}}} \left[\left\| f_{x_i} - f_{x'_i}\right\|_2\right] \tag{13}$$

$$\mathcal{L}_{f_{\text{aug}}} = \mathbb{E}_{x_{\text{aug}}, y \sim P'_{\mathcal{D}}(x_{\text{aug}})} \left[\left\| f_{x_i} - f_{x_{\text{aug}}}\right\|_2\right] \tag{14}$$

$$\mathcal{L}_f = \mathcal{L}_{f_i} + \mathcal{L}_{f_{\text{aug}}}, \quad i = 1, 2. \tag{15}$$

To facilitate discriminative representation learning, augmented samples should be more challenging than original samples. The regularization loss proposed by Li et al. [48] is adopted to maximize the difference between the cross-entropy losses on training samples x_i and augmented samples x_{aug}, which is calculated as follows:

$$\mathcal{L}_c^{\text{reg}} = \left|1.0 - \exp\left[\mathcal{L}_c(x_i) - \rho \mathcal{L}_c^{\text{aug}}\left(x_{\text{aug}}\right)\right]\right| \tag{16}$$

where $\rho = \max(1, \exp(\sum \hat{y} \cdot y))$ $(\rho \geq 1)$ is a dynamic parameter to control the gradual generation of challenging augmented samples in the training process, and $y \in [0, 1]$ is the category prediction of x_i and x_{aug}.

In this article, we propose to perform a view semantic transformation that produces the "hallucinating" examples to balance the embedding of target images in the feature space to approach the actual SAR data distribution. However, the two-stage learning approach of synthetic samples and target recognition is suboptimal for two reasons. First, generating high-quality synthetic samples does not equal the gain in the performance of recognition tasks. Second, high-resolution SAR images generated by a generative model might still fail to capture the many modes of variation of visual concepts due to mode collapse. Therefore, we propose simultaneously performing SAR image synthesis and SAR target recognition. As shown in Fig. 4, the joint learning strategy obtains a better

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decision boundary. On the one hand, this design effectively reduces the search scope of the augmented generation through end-to-end optimization. On the other hand, the representation learning task limit and guide the solution space of the semantic augmentation module to avoid unnecessarily searching for more accurate or complex solutions.

Specifically, we adopt a two-stage alternating training process to optimize the model's parameters. In the first stage, the learning process of the model is to calibrate the biased distribution at sparse views through view propagation. In the second stage, we train the recognition model based on the calibrated distribution that can be generalized to a wide range of target recognition scenarios. Therefore, the proposed VSTNet overcomes the distribution shift and significantly improves the performance of SAR target recognition.

Finally, we amalgamate the SAR image synthetic and SAR target recognition into a unified, end-to-end framework. The overall objective function is expressed as follows:

$$\mathcal{L}^{\text{inner}} = \left[\mathcal{L}_d, \mathcal{L}_g, \mathcal{L}_r + \mathcal{L}_{\text{div}} + \mathcal{L}_f\right] \tag{17}$$

$$\mathcal{L}^{\text{outer}} = \mathcal{L}_c^{\text{reg}} + \mathcal{L}_c + \mathcal{L}_c^{\text{aug}} \tag{18}$$

where $\mathcal{L}^{\text{inner}}$ is the overall loss for image synthesis and $\mathcal{L}^{\text{outer}}$ is the overall classification loss.

D. Framework Discussion

SAR target recognition with sparse views can eliminate the view dependence and improve the practicality of the deep-learning model. We propose a VSTNet to simultaneously perform SAR image synthesis and SAR target recognition. Compared with the analysis in Section I, this section presents an intuitive discussion to highlight the superiority further.

Current methods for SAR target recognition often focus on small samples and have attempted to increase training data via image generation or data augmentation. However, these methods have focused solely on sample synthesis and have yet to consider optimization simultaneously. The separation of data synthesis and target recognition imposes a significant resource burden on practical applications. Therefore, the VSTNet proposed in this article would amalgamate distribution calibration and discriminant feature learning into a unified end-to-end optimization framework that effectively supplements existing target recognition methods.

Many methods can transfer prior knowledge from optical image data, simulated SAR data, or pretrained networks from the source domain to the target domain. However, the differences in target pose, scene illumination, and background clutter between the source domain and the target domain lead to data distribution differences between the two fields. The biased distribution caused by the limited views focused on in this article, combined with the above distribution difference, will lead to a negative transfer. This worse phenomenon increases the challenge of target recognition, and blindly borrowing knowledge from other fields makes the results of target recognition ambiguous. Therefore, the proposed VSTNet avoids the above disadvantages by mining and utilizing the characteristics of the target image to improve the performance of target recognition.

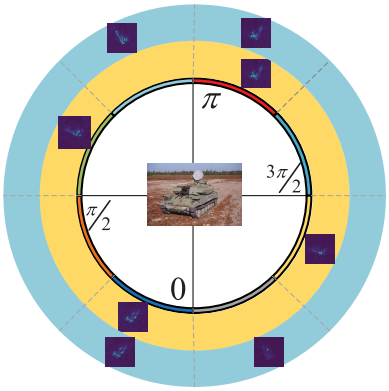
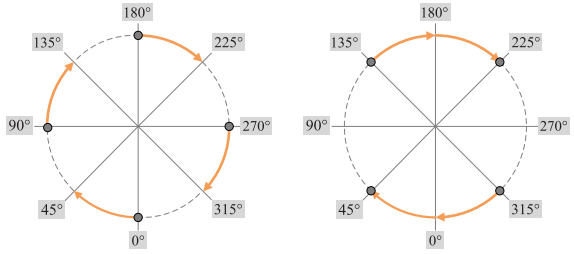


Fig. 5. Illustrated SAR sensor system operates in a different mode. (a) Red ring: the interval between sparse views is uniform and with a minimum of 45°. (b) Green ring: the interval between sparse views is nonuniform and with a minimum of 90°.



(a) (b)

Fig. 6. Two scenarios of incomplete data caused by the SAR system working in different modes. (a) Interval of views is uniform in MSTAR. (b) Interval of views is non-uniform in MSTAR.

Multiview learning utilizes complementary information and internal correlation between the target images at different aspect angles. However, due to the special imaging mechanism of SAR, the characteristics of the target image will become discrete as the interval of aspect angle increases. When the interval of the aspect angle is large, the poor performance of the existing multiview learning methods integrated with sparse representation should be related to the SAR imaging mechanism. We take the aspect angle into account in the target generation model, which has an important influence on SAR interpretation. Therefore, the proposed VSTNet calibrate the biased distribution with samples at limited aspect angles because of combining the SAR physical dimension information. In this way, the recognition model trained with inputs sampled from the calibrated distribution can improve the performance of the target recognition.

IV. Experiments and Analysis

In this section, we evaluate the effectiveness of the proposed VSTNet with a series of experiments. The proposed method aims to unify SAR image synthesis and SAR target recognition into an end-to-end framework to boost their performance against each other. First, we introduce the dataset used in our experiments. Then, we use a series of ablation experiments to verify the performance of the critical modules of VSTNet. Finally, the VSTNet will be compared with other state-of-the-art methods to demonstrate its effectiveness further.

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TABLE I
TEN TYPES OF SAR TARGET DATA IN SPARSE VIEW

		2S1	BMP2	BRDM2	BTR60	BTR_70	D7	T62	T72	ZIL131	ZSU234
Train (17°)	(a)	165	114	160	127	126	135	158	119	161	160
	(b)	156	115	156	123	112	148	156	116	156	155
Test (15°)		274	196	274	195	196	274	273	196	274	274

TABLE II
THREE TYPES OF SAR TARGET DATA IN SPARSE VIEW

		2S1	BRDM2	ZSU234
Train (17°)	(a)	165	160	160
	(b)	156	156	155
Test	30°	288	287	288
	45°	303	303	303

TABLE III
STRUCTURES OF THE ENCODER AND THE GENERATOR

Encoder Input:(64×64×1)		*Generator* Input:(1×1×$views$, 1×1×$classes$)	
Conv1+BN+Relu	32×32×64	Deconv1+BN+Relu	4×4×1024
Conv2+BN+Relu	16×16×128	Deconv2+BN+Relu	8×8×512
Conv3+BN+Relu	8×8×256	Deconv3+BN+Relu	16×16×124
Conv4+BN+Relu	4×4×512	Deconv4+BN+Relu	32×32×64
Conv5	1×1×$views$	Deconv5	64×64×1

A. Datasets for Experiments

1) Data Description: The moving and stationary target acquisition and recognition (MSTAR) public dataset is widely used to verify the effectiveness of the proposed method. It was collected by the Sandia National Laboratory SAR sensor platform, which was sponsored by the Defense Advanced Research Laboratory [49].

We utilize two application scenarios for SAR sensors with sparse views to verify the effectiveness of the proposed method, as shown in Fig. 5. The distributions of the sparse views in the two scenarios are as follows: 1) the interval of views is uniform and 2) the interval of views is nonuniform. We selected the following view ranges from the MSTAR, as shown in Fig. 6, to obtain the two incomplete datasets: 1) [0°, 45°], [90°, 125°], [180°, 225°], [270°, 315°); and 2) [0°, 45°], [135°, 180°], [180°, 225°], [315°, 360°). Objectively, the minimum interval between sparse views is 45° or 90°, and scenario Fig. 6(b) is more complicated than scenario Fig. 6(a).

2) Data Preprocessing: To reduce redundant background and the computational cost, we crop SAR images to a 64 × 64 pixels region of interest, which is commonly used to preprocess the images of the MSTAR dataset.

This article focuses on two experimental scenarios: ten-classes SAR target recognition and three-classes SAR target recognition. The real-world scenarios presented in Tables I and II comprehensively explain the previously discussed experimental environment. From Table I, ten types of targets with a depression angle of 17° have been selected for training, and those with a depression angle of 15° have been selected for testing. As indicated in Table II, images of 2S1, BRDM2, and ZSU234 targets at a depression angle of 17° have been chosen for training, while images at depression angles of 30° and 45° are designated for testing.

B. Training Detail

The proposed VSTNet consists of four subnetworks. The structures of the encoder E and generator G are shown in Table III, and the input and output are specified in detail. The discriminator D has a simple structure of three fully connected layers followed by a ReLU activation function with 128, 32, and 1 units. The VGG16 is chosen as the classifier structure. To coordinate the SAR image synthesis and SAR target recognition, we use five optimizers: $\mathrm{lr}_{\mathrm{ae}} = 5e-4$, $\mathrm{lr}_d = 5e-4$, $\mathrm{lr}_g = 5e-4$, and $\mathrm{lr}_c = 1e-4$. In this experiment, we chose the value of views to be 9. The parameter settings of the classification regularity are consistent with the literature [48]. The experiments are implemented using the Tensorflow deep-learning framework, and all experiments are conducted on an NVIDIA GTX 2080Ti GPU.

C. Ablation Experiments

In this section, we design experiments to demonstrate the effectiveness of the proposed method and show the recognition results of ablation experiments in Table IV. We highlight the advantages of the proposed method, which are attributed to two key components: the semantic transformer network and joint representation learning. The former is primarily concerned with the rationalization of the prior and the validity of semantic transformations. In particular, several variants are considered.

1) *Variant-1:* A variational autoencoder model with the normal distribution prior.
2) *Variant-2:* A simple disentanglement model whose structure is the same as an adversarial autoencoder.
3) *Variant-3:* Multitask learning replaces the manner of task feedback optimization.

1) Effectiveness of View-Semantic Transformer: According to Fig. 6, the incompleteness of the training data caused an under-representation of the model and reduced recognition performance. When this occurs, the model's generalizability to unknown views primarily determines how well it performs. Therefore, we evaluate the model's performance by combining two aspects: the views included in the training set and those

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TABLE IV
RESULTS OF ABLATION EXPERIMENTS FOR SCENARIO (b)

Algorithm	Accuracy (%)		
	accuracy_in	accuracy_ex	mean±std
Variant-1	98.55	56.34	77.98±0.26
Variant-2	99.10	59.37	79.94±0.47
Variant-3	98.87	53.85	76.94±1.89
VSTNet	**99.21**	**70.31**	**85.82±0.13**

extended outside the training set, which are simply noted as "accuracy_in" and "accuracy_ex."

From the comparison results in Table IV, it is clear that the poor performance of Variant-1 makes the normal prior unsuitable. Adopting an adversarial strategy to impose a uniform distribution on the hidden space using discriminators is necessary, which avoids the complexity of KL divergence. The initial prototype of the proposed method, that is, Variant-2, is then obtained. Also, the proposed method significantly improves generalization over Variant-2, which shows the effectiveness of the semantic transformation module. The proposed VSTNet uses the fact that features in the latent space are separated linearly and translates category-related features along the manifold directions of the views. Therefore, the purpose of the semantic transformation module is to ensure that the translated features have the same category but different view semantics than the original. The model fills in the location of the target in the feature space for unknown views by transforming the feature space, as shown in Fig. 4. Finally, the corrected and completed biased data distribution is also guaranteed for the target recognition module to obtain better decision boundaries, which greatly improves the generalization performance for unknown views.

2) Effectiveness of Task Feedback Optimization Strategy: By comparing the results from Table IV, both the proposed method and Variant-2 perform significantly better than Variant-3 on ten-classes of targets, which makes it necessary to adopt suitable techniques to nest the data synthesis and recognition tasks so that they can reach their full potential. For traditional strategies with multiple losses happening simultaneously, these losses may cancel each other out, leading to a failure to improve model performance. We are inspired by the idea that complementary knowledge might flow from one task to another. Hence, the VSTNet uses joint representation learning to coordinate semantic augmentation generation and discriminative representation learning. This way, the two subtasks can be done one after the other while helping each other until convergence.

Finally, the results of the ablation experiments shown in Table IV confirm that the semantic transformer network and joint representation learning are effective, and the proposed VSTNet is beneficial for alleviating the sensitivity of SAR target recognition to sparse views.

D. Framework Comparison for SAR Target Recognition

To further verify the performance of the proposed method, it is compared with some related methods: traditional methods, for example, the k-nearest neighbors algorithm (K-NN) and SRC [50]; CNN-based methods, for example, AconvNet [51]; synthetic data-based methods, for example, GAN; transfer learning-based methods, for example, VGG16 [52] and ResNet50 [53]; multiview learning-based methods [20]. The results of the VSTNet method on the ten-classes of SAR targets for two application scenarios are shown in Tables V and VI. In addition, Fig. 7 shows the performance of the proposed method for three-classes of SAR targets at different depression angles.

TABLE V
COMPARISON OF RECOGNITION ACCURACY (%) ON TEN-CLASSES OF TARGETS FOR SCENARIO (a)

Algorithm	Accuracy		
	accuracy_in	accuracy_ex	mean±std
AconvNet	96.85	72.58	85.28±0.59
GAN+CNN	96.06	80.19	89.91±0.55
VGG16	96.27	77.02	87.10±0.32
ResNet50	97.87	76.36	87.62±0.61
STN	97.37	78.92	88.58±0.90
KNN	88.42	42.27	66.43±0.22
SRC	94.32	44.18	62.00±0.01
4VDCNN	96.85	72.32	85.41±0.18
VSTNet	**98.54**	**85.62**	**94.45±0.23**

TABLE VI
COMPARISON OF RECOGNITION ACCURACY (%) ON TEN-CLASSES OF TARGETS FOR SCENARIO (b)

Algorithm	Accuracy		
	accuracy_in	accuracy_ex	mean±std
AconvNet	97.49	43.96	71.41±0.87
GAN+CNN	93.24	63.27	79.12±4.30
VGG16	97.67	56.34	77.53±0.37
ResNet50	94.55	51.88	73.76±0.93
STN	97.66	48.03	73.48±0.90
KNN	92.07	37.73	65.60±0.23
SRC	93.30	35.03	55.98±0.21
4VDCNN	95.09	46.15	71.25±0.37
VSTNet	**99.21**	**70.31**	**85.82±0.13**

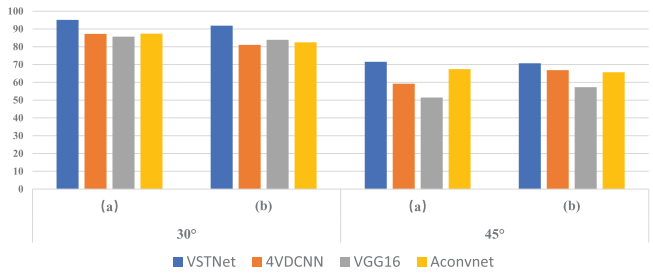


Fig. 7. Recognition accuracy (%) of classical methods. (a) Interval of views is uniform on the 3-type dataset. (b) Interval of views is non-uniform on the 3-type dataset.

As can be seen from Tables V and VI, the designed VSTNet model can achieve very competitive experimental results for multiple classes of targets. First, the proposed method uses the idea of data synthesis to correct the biased distribution of data in feature space. As shown by "accuracy_ex" in

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Tables V and VI, this method significantly alleviates the under-representation of the target recognition model. Second, joint representation learning is used to guide the data synthesis to converge toward improved recognition performance by training sub-modules alternatively. In addition, as shown in Fig. 7, we validated the method on three classes of SAR target data to demonstrate its robustness. Therefore, the proposed method breaks the dependency of SAR target recognition on complete views and generalizes open application scenarios.

V. Conclusion

In this article, we propose the VSTNet to improve the performance of SAR images in sparse views. The proposed VSTNet first designs a view semantic augmentation module to generate augmented samples to complement the biased distribution of sparse-view data. The aim is to balance the embedding of target images in the feature space to approach the distribution of complete SAR data. Second, the recognition model is trained with samples obtained from the augmented feature space and can generalize well. Finally, we design a unified framework to amalgamate semantic feature augmentation and discriminant feature learning into an end-to-end optimization framework. In addition, this framework takes discriminant representation learning as the main task that constrains the solution space of the augmented generation task and guides it to generate features that favor the optimal decision boundary, thus avoiding more complicated solutions. The proposed method is not limited by the continuity of view and is oriented toward more general application scenarios. Extensive experiments have been conducted on the MSTAR dataset, where some state-of-the-art methods are employed as competitors. The experimental results have demonstrated the robustness and superiority of the proposed method for SAR target recognition. In the future, we will consider combining the physical imaging characteristics of SAR to improve the performance of target recognition with incomplete data.

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